

AI & API Operational Costs Analysis

Prepared For	Prepared On	Status
Inventory System Project	June 2, 2026	Approved Production Spec

This document details the breakdown of operational costs associated with our search, recognition, and valuation pipelines. It delineates local, zero-cost operations from metered third-party cloud API services, establishing the baseline cost structure for each user action.

1. Local & Free Processing (\$0 Cost)

To optimize overall cost margins, several heavy data-processing tasks run client-side on the user’s mobile hardware or hit unmetered public API registries:

- **On-Device Barcode Recognition:** Decodes UPC-A, UPC-E, EAN-13, EAN-8, QR, Code-128, etc., directly on the mobile CPU/GPU using native libraries via the expo-camera framework.
- **Image Quality Compression:** The scanner app automatically compresses full-resolution images down to a quality ratio of 60% (~400KB files) prior to server sync, protecting local bandwidth.
- **Local Database Engine:** All asset relationships, user configurations, sync queues, and caching are managed offline in local SQLite databases on both mobile and native desktop apps.
- **Public Registry Lookups:**
 - **Google Books API:** Barcodes matching ISBN books (starting with 978 or 979) query Google Books without charge or key requirements.
 - **Open Food Facts API:** Decodes standard food products via an open, crowdsourced database.
 - **UPCItemDB Trial Tier:** fallback search engine for standard product codes (free trial mode, capped at 100 requests/day).

2. Paid Cloud Services & Billed Actions

Billed calls occur during background sync processing (orchestrated by `lib/worker.js`) whenever a user scans collectibles or requires premium enrichment:

- **Basic Photo Identification (Google Cloud Vision API):** Triggered for standard image imports. Runs text recognition, web matching, labels, and logo checks.
- **Deep Visual Matching (SerpApi Google Lens):** Runs when the user activates "Premium Tier" or scans specific categories. Re-uploads the image and queries Google Lens for visually matching items and exact titles.
- **Market Valuation Pricing (SerpApi Google Shopping):** Triggered for collectible appraisal (vintage toys, coins, cards, and comics). Returns pricing ranges by analyzing shopping listings.
- **Slab/Grading Cert Validation (SerpApi Organic Search):** Triggered when a graded slab barcode is detected. Queries Google to scrape certificates from PCGS, NGC, PSA, BGS, SGC, or CGC.
- **Entertainment Metadata Enrichment (SerpApi Search & YouTube):** Triggered when scanning Movies and TV shows. Resolves IMDb organic search snippets for plot/cast lists and retrieves YouTube links for official trailers.

3. API Models, Pricing, and Cost Breakdown

Important Note on Token-Based Billing

No large language models (such as GPT-4 or Gemini Multimodal API) are utilized for these pipelines. All data queries are based on structured computer vision and search engine integrations. There are no input/output token counts or token-based costs.

Service & Endpoint	Integration Method	Billing Model	Cost per Call
Google Cloud Vision	v1/images:annotate	4 Features Billed per Image (Text, Logo, Web, Labels)	\$0.0085

Service & Endpoint	Integration Method	Billing Model	Cost per Call
SerpApi (Google Lens)	engine=google_lens	1 Search Credit per Lookup	\$0.0100
SerpApi (Google Shopping)	engine=google_shopping	1 Search Credit per Lookup	\$0.0100
SerpApi (Search & YouTube)	engine=google / youtube	1 Search Credit per Lookup	\$0.0100
Numista Coin API	/api/v3/types	Billed per Subscription (Custom)	Variable

4. Cost Per User Action Scenarios

Standard Barcode Scan

\$0.00

User scans a standard retail item barcode. Resolved locally and matched against Google Books, Open Food Facts, or trial registries.

Basic Photo Scan

\$0.0085

User takes a photo of a standard item in basic mode. Calls Google Cloud Vision to detect basic tags, logos, and text.

Premium Image Match

\$0.0185

User takes a photo in Premium Mode. Google Vision text OCR resolves basic text, and SerpApi Google Lens finds visual matches.

Collectible Valuation

\$0.0385

Premium scan of a coin, trading card, or vintage toy. Performs Google Vision OCR, SerpApi Google Lens lookup, SerpApi Google Shopping search, and search-based cert validation.